The Positive-Chamber Koide Identity for Icosahedral Face Circulants

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Abstract

The stabilizer of an oriented icosahedral face acts regularly on its three corners. Every Hermitian response commuting with this cyclic action is a circulant $C = aI + bR + \bar{b}R^2$, $R^3 = I$. In the chamber where all three eigenvalues are nonnegative and may be read as square-root masses, its Koide invariant is

$$Q = \frac{1}{3} + \frac{2}{3}\left(\frac{|b|}{a}\right)^2.$$

Thus $Q = 2/3$ is exactly equivalent to the single balance condition $|b|/a = 1/\sqrt{2}$. This identity is a mathematical result about positive C_3 face circulants. The phase of b drops out of Q and controls the two mass ratios jointly; it is supplied, not derived. A finite tracial Gelfand–Naimark–Segal construction that pairs the neutral line with a two-state orientation record and the charged plane with one oriented event gives the balanced modulus as a conditional theorem. The balance itself is a target-informed premise: it is chosen from the measured charged-lepton triple and is algebraically the relation Koide published. Under the balance and mass-ordering premises the measured electron and muon masses fix the tau mass inside a 72-eV enclosure centered at 1776.969027 MeV. The world average lies 0.43 and Belle II 2023 lies 0.89 of their standard uncertainties away; both comparisons are retrospective, and a registered decision rule with a 0.045 MeV precision floor governs later measurements. Koide's relation gives the same mass for every input, so their likelihood ratio is one. The local face fiber has no constructed physical chiral-family attachment, so the result is a representation-theoretic identity, a finite conditional implication and a conditional tau-mass window; no source-only mass prediction is claimed.

Keywords: Koide relation, circulant matrices, cyclic symmetry, icosahedral symmetry, finite event algebra, GNS representation

One theory and its executable hypothesis. OPH proposes closure: one complete history is both a timeless self-referential structure and its own internal computation, and observers inside it recover and realize its specification. The hypothesis that makes this proposal concrete and testable is executable.

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A simple, fully specified local repair and memory law on the proposed carrier architecture admits continuing observers whose compatible records and interventions have the observed $3+1$-dimensional quantum-field description and a common classical-gravity limit. The preferred matter mechanism is localized topological defects of native overlap transport, with rest energy derived from their excess maintenance dynamics in the same physical clock and energy dictionary.

One source process must meet every part of this statement: records, clocks, matter, energy and responsive geometry come from the same law, state and refinement. Coherence, protected codes, radial directions, fields, clocks and couplings may be specified as part of its design; their implementation and observable consequences must be established on that process. A failed specified law rejects that law and its tested interpretation; it rejects the wider proposal only when the rejected choice is proved necessary.

The tower. The papers form one tower over the executable model. Each layer states a hypothesis and proves finite theorems under displayed premises. A conclusion passes to the next layer only after that premise is proved on the same model process; until then it is a conditional implication.

0. **Closure and agreement.** One history reads itself into existence, and agreement on shared records defines the public world. This is the interpretation and acceptance criterion; no theorem selects a unique universe.

1. **Repair mathematics.** Local agreement repair reaches global consistency, and nonzero cycle holonomy is the residue that local repair cannot remove. Fixed-point, normal-form, frustration and finite event-algebra theorems hold on their stated domains.

2. **Carrier and repair law.** Federated twelve-port icosahedral carriers with the native C/E seam repair are the substrate. The seam rule follows from four stated requirements, the quantum record laws are classified, and the uniform native repair selects a rank-three slow-response band.

3. **Executable observer.** Observers are continuing native processes that prepare, read, retain and act on records. Finite complete-output executions, overlapping-observer correspondence, event clocks and checkpoint continuation are verified; one continuing observer with every service and resource accounted is the open join.

4. **Clocks and fields.** Modular flow of observer algebras supplies candidate local clocks, and native repair supplies a positive continuum generator of spectral dimension two on the screen. A declared radial extension gives a conditional smooth $1+3$ scalar description. Complete finite source factors join spatial CAR dynamics, particle creation, repair and a common calibrated clock. Physical validation of that clock and energy normalization, and an interacting chiral $3+1$ limit, remain open.

5. **Gauge structure and matter.** Gauge structure is the transport of overlap data, and stable massive matter is a localized defect whose boundary transport carries a holonomy that admitted repairs cannot erase. The compact Lie type $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$ follows under complete-response premises, and fixed nonzero holonomy forces positive residual mismatch. A specified coherent ordering memory protects native exchange-history sectors, and a colored

extension implements their fermionic exchange. A localized physical defect, its maintenance-to-mass law and a predicted measured spectrum remain unestablished; the earlier double-load candidate is transient.

6. **Gravity.** Einstein dynamics is the coarse response of the same process. The null-balance and tensor-completion implication is proved under its premises; realizing those premises on one refinement tower is open, and the static radial product fails null balance.

7. **Cosmology and tests.** The same process must fix the vacuum, dark sector, primordial response and black-hole records, and pass independent prospective tests. The cosmology papers and the fine-structure, Koide, de Sitter and Yang–Mills supplements carry conditional branches, diagnostics and retrospective comparisons at this layer.

Completion and stopping. The aim is a consistent and complete theory: every layer realized on one process, or a proved obstruction that removes OPH or a stated branch of it. Results that no join needs leave the active chain with their provenance. A universal obstruction over a frozen admissible class, an unavoidable excluded prediction, or a contradiction among indispensable premises stops that class. A failed candidate, an unconstructed map or a resource stop has a narrower meaning. The ordered joins, retained foundations and stopping rules are recorded with the executable model in the public repository.[1]

Fixed physical boundaries. Repair counts, host runtime and entropy production are distinct from energy; the maintenance-to-energy map must be derived with one common signal speed and no per-species conversion. The modular parameter is dimensionless until a specified and implemented clock calibration relates it to duration. Stationary repair or record entropy is distinct from the fixed-volume generalized-entropy variation that the Einstein construction consumes. A massless photon makes the infinite-volume theory gapless, so sector gaps and thresholds name their state and limit; confinement precludes free massless gluons as observable particles; a quantum graviton lies outside the classical Einstein theorem.

This paper's hypothesis and place in the tower

This paper is a layer-seven supplement to the matter layer. It proves an unconditional finite identity: for a positive Hermitian C_3 face circulant, the Koide invariant depends only on the modulus ratio $|b|/a$, and $Q = 2/3$ is exactly $|b|/a = 1/\sqrt{2}$. Its physical hypothesis is narrower and stated as a premise: the charged-lepton square-root masses are the positive eigenvalues of one balanced face circulant. The balance is target-informed, since it is read off the measured charged-lepton triple, and the phase that fixes the two mass ratios is supplied. The finite tracial theorem shows one event packet that realizes the balance; its asymmetric block choice carries that premise.

For any physical reading the executable model must realize what the theorems take as input: a native localized defect sector carrying three chiral charged families, a quotient-visible map from the local face fiber to that family response, preservation of the block powers, a derived phase, and a rest-energy readout in the common clock and energy dictionary. None of these is constructed. A tau-mass measurement that meets the registered precision floor and lies more than three standard

[1] https://github.com/FloatingPragma/observer-patch-holography/blob/r2058/docs/COMPLETION_PROGRAM.md

uncertainties from the window center refutes the balanced-circulant premise; it leaves the identity untouched.

1 The face-corner carrier

Koide's charged-lepton relation [1] is usually written in terms of the three positive square-root masses. The construction below identifies the exact condition it imposes on a cyclic Hermitian response.

Let $G = A_5$, the alternating group on five letters, act by proper rotations on the icosahedron. The twenty outward oriented faces form the transitive orbit G/C_3. The stabilizer of one face cyclically permutes its three corners, so every face carries a local copy of the regular C_3 representation. If R denotes the cyclic shift, then

$$R^3 = I, \qquad R^\dagger = R^2.$$

The commutant of this regular action is the three-dimensional circulant algebra

$$\{x_0 I + x_1 R + x_2 R^2 : x_j \in \mathbb{C}\}.$$

Hermiticity restricts a response to

$$C = aI + bR + \bar{b}R^2, \qquad a \in \mathbb{R}, \quad b \in \mathbb{C}. \tag{1}$$

Write $b = \rho e^{i\delta}$, with $\rho = |b| \geq 0$. Fourier diagonalization gives

$$\lambda_k = a + 2\rho \cos\left(\delta + \frac{2\pi k}{3}\right), \qquad k = 0, 1, 2. \tag{2}$$

The unordered spectrum is independent of the chosen face representative. This is a local bundle statement. The sixty face-corner flags form the regular A_5 torsor; the geometry alone supplies no canonical global three-dimensional physical family space.

2 The positive-chamber identity

Theorem 1 (Positive-chamber Koide identity)**.** *Let C be the Hermitian circulant in Eq. (1), with $a > 0$. Suppose its three eigenvalues in Eq. (2) are nonnegative and, for one $s > 0$,*

$$\sqrt{m_k} = \sqrt{s}\lambda_k.$$

Then

$$Q := \frac{\sum_{k=0}^{2} m_k}{\left(\sum_{k=0}^{2} \sqrt{m_k}\right)^2} = \frac{1}{3} + \frac{2}{3}\left(\frac{\rho}{a}\right)^2. \tag{3}$$

Consequently,

$$Q = \frac{2}{3} \quad \Longleftrightarrow \quad \frac{\rho}{a} = \frac{1}{\sqrt{2}}. \tag{4}$$

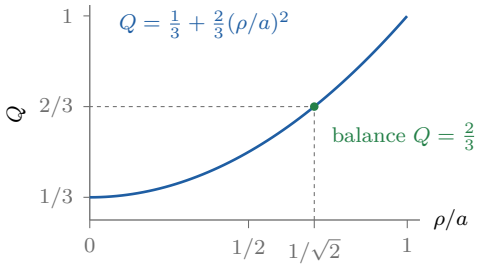
Proof. Set $c_k = \cos(\delta + 2\pi k/3)$. The roots-of-unity identities give

$$\sum_{k=0}^{2} c_k = 0, \qquad \sum_{k=0}^{2} c_k^2 = \frac{3}{2}.$$

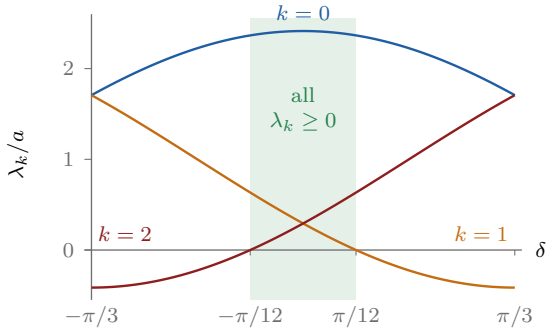
It follows that

$$\sum_k \lambda_k = 3a, \qquad \sum_k \lambda_k^2 = 3a^2 + 6\rho^2.$$

The scale s cancels from Q, yielding Eq. (3). Since $\rho/a \geq 0$, Eq. (3) equals $2/3$ exactly at $\rho/a = 1/\sqrt{2}$. □



(a) Koide ratio when $\sqrt{m_k} \propto \lambda_k \geq 0$



(b) $\lambda_k/a = 1 + \sqrt{2}\cos(\delta + 2\pi k/3)$ at balance; the positive chamber is $|\delta| \leq \pi/12$ modulo $2\pi/3$

Figure 1: (a) The Koide ratio of a nonnegative spectrum of $C = aI + bR + \bar{b}R^2$, $\rho = |b|$, as a function of ρ/a (Theorem 1); balance $\rho/a = 1/\sqrt{2}$ gives $Q = 2/3$. (b) At balance the three eigenvalues are nonnegative exactly in the chamber $|\delta| \leq \pi/12$ modulo $2\pi/3$, the chamber outside which the theorem makes no statement.

Remark 2 (The chamber is part of the theorem). The signed trace calculation holds algebraically for every δ. Physical square roots use $|\lambda_k|$ when an eigenvalue is negative, and then the denominator is not the signed trace. At $\rho/a = 1/\sqrt{2}$, the positive chamber is

$$|\delta| \leq \frac{\pi}{12} \pmod{\frac{2\pi}{3}}.$$

Theorem 1 makes no physical Koide statement outside that chamber.

Remark 3 (What the phase contains). Inside the positive chamber, Q is independent of δ. The three eigenvalues, and hence the two reported mass ratios, vary with δ. The balance condition removes one scale-free degree of freedom from a three-mass spectrum. It does not determine the ratios.

3 Conditional finite tracial balance

The balance condition in Eq. (4) can arise exactly from a finite event packet. This section states the packet as an independent conditional theorem. The packet realizes the target-informed balance premise through an asymmetric block choice, which is where that premise enters.

Let

$$\mathcal{V} = \mathbf{1} \oplus \chi \oplus \overline{\chi}, \qquad \mathcal{H}_{\mathrm{or}} = \mathbb{C}^2, \qquad \mathcal{A} = B(\mathcal{V} \otimes \mathcal{H}_{\mathrm{or}}) \simeq M_6(\mathbb{C}).$$

Here $\mathbf{1}$ is the neutral cyclic mode, $\chi \oplus \overline{\chi}$ is the two-dimensional charged plane, and $\mathcal{H}_{\mathrm{or}}$ is a two-state orientation record. Let P_0 and P_c project onto the neutral line and charged plane. For a rank-one oriented event q_+, define

$$Z_0 = P_0 \otimes I_{\mathrm{or}}, \qquad Z_c = P_c \otimes q_+, \qquad E_+ = Z_0 + Z_c.$$

Both Z_0 and Z_c have rank two.

Theorem 4 (Conditional finite tracial-GNS balance)**.** *Condition the normalized trace state $I_6/6$ on E_+. Map the orthonormal cyclic basis of the resulting square-root amplitude to (I, R, R^2) in $L^2(M_3(\mathbb{C}), \tau_3)$. Then*

$$p_0 = p_c = \frac{1}{2}, \qquad a = \sqrt{p_0}, \qquad \rho = \sqrt{\frac{p_c}{2}},$$

and therefore

$$\frac{\rho}{a} = \frac{1}{\sqrt{2}}, \qquad Q = \frac{2}{3}$$

whenever the positive square-root-mass reading of Theorem 1 is supplied.

Proof. Normalized-trace conditioning assigns probability proportional to block rank. Both accepted blocks have rank two, so $p_0 = p_c = 1/2$. In $L^2(M_3(\mathbb{C}), \tau_3)$, the operators I, R, R^2 are orthonormal. The canonical square-root vector therefore has neutral coefficient $\sqrt{p_0}$ and two charged coefficients of modulus $\sqrt{p_c/2}$. Their ratio is $1/\sqrt{2}$. Theorem 1 then gives $Q = 2/3$ on the positive chamber. □

Theorem 4 is finite and exact. Its event algebra, block choice, conditioned trace, and cyclic response map are premises. The block choice is asymmetric: the neutral line is paired with the full two-state orientation record and the charged plane with one oriented event. Treating the two blocks alike, both with I_{or} or both with q_+, gives $p_c/p_0 = 2$, hence $\rho/a = 1$ and $Q = 1$. The normalized trace fixes the probabilities only after the asymmetric event has been chosen, so the balanced modulus is the content of that choice. A physical charged-lepton statement additionally requires a quotient-visible map from the local face bundle to one chiral three-family response, preservation of the relevant block powers, and a mass readout. Those objects are absent from the theorem. The executable observer construction supplies none of them: the tau window consumes no native observer run, and its identification of balanced square-root masses with charged-lepton pole masses is a premise [7, 6].

Remark 5 (What a physical attachment would add). The connected register fixes the normalization left free by a direct-sum maximum-entropy model; minimality of this response-local register is a declared hypothesis. A physical charged-lepton conclusion requires a chiral regular-C_3 carrier and faithful trace-preserving unital completely positive maps Φ, Ψ between the source and physical response algebras, with $\Psi\Phi = \mathrm{id}$. If they intertwine the block records and identify the physical GNS vector with $C = (Y_e^\dagger Y_e)^{1/4}$, Kadison–Schwarz makes Φ a $*$-monomorphism and an exact L^2 isometry, so it carries the equal source block powers to the physical response. For an accepted finite chiral checkpoint with three left and three right family modes, positive kinetic metrics, and a committed neutral Higgs direction of electroweak scale v, $\Phi = \mathrm{Ad}_{J_L \oplus J_E}$. With $M_F = X_F^2$, this gives

$$\widehat{Y}_e = \frac{\sqrt{2}}{v} J_L M_F J_E^\dagger, \qquad \mathcal{M}_L = J_L M_F J_L^\dagger.$$

An extended branch that adds graded physical completion and quotient source-law selection selects one dressed mass readout, given an exhaustive declared carrier class with a positive winner gap, a source-closed BV/BRST continuum, and an interval or contraction certificate for the charged QFT self-map. A balanced C_3 fixed point then gives $Q = 2/3$; a C_3-symmetric fixed point with charged attenuation $\chi_\star$ gives

$$Q = \frac{1 + e^{-2\chi_\star}}{3}.$$

An off-plane response requires the full response operator. None of these maps, checkpoints or certificates is constructed.

4 The conditional tau-mass window

Fixing $Q = 2/3$ in Eq. (3) and inserting the measured electron and muon masses turns the Koide coordinate into one quadratic in $\sqrt{m_\tau}$. The mass-ordering premise $m_\tau > m_\mu$ selects the physical root; the excluded second root sits at 3.317 MeV, below the muon mass. Propagating the one-sigma corners of the measured (m_e, m_μ) inputs [2] with outward rounding encloses the selected root in

$$m_\tau \in [1776.968991,\ 1776.969063]\ \text{MeV}, \tag{5}$$

a 72-eV window with center 1776.969027 MeV. The center sits 0.43 standard uncertainties from the measured 1776.93 ± 0.09 MeV, and the window is three orders of magnitude narrower than that measurement uncertainty. The dedicated Belle II 2023 measurement, $1777.09 \pm 0.08_{\text{stat}} \pm 0.11_{\text{syst}}$ MeV, lies 0.12097 MeV above the center, 0.89 of its combined 0.136 MeV uncertainty [4]. That comparison is retrospective, and its systematic uncertainty alone exceeds the 0.045 MeV compatibility floor, so it carries no verdict under the registered rule [5].

The comparison is governed by a registered conditional test whose target definition and decision rule were fixed in a frozen prediction record [3]: a world-average or dedicated-measurement central value more than three standard uncertainties from 1776.969027 MeV refutes the balanced-circulant premise; compatibility requires agreement within two standard uncertainties at a measurement uncertainty of at most 0.045 MeV; every other outcome carries no verdict. The balance premise, the equal-weight condition on the two response sectors of one icosahedral face circulant, is target-informed, so the window is a conditional postdiction whose evidential weight lies in the kill direction [3, 6]. Improving tau-mass measurements test the balanced-circulant premise and Koide's relation together. Both send every allowed (m_e, m_μ) to the same m_τ, so under one measurement model their likelihood ratio is one and no precision in this observable distinguishes them; the formula contains no screen-grain coordinate P and constrains no other OPH parameter [6].

5 Numerical diagnostic and provenance

Using the Particle Data Group 2026 central masses [2],

$$(m_e, m_\mu, m_\tau) = (0.51099895069,\ 105.6583755,\ 1776.93)\ \text{MeV},$$

gives

$$Q_{\text{PDG}} = 0.6666644634026367.$$

The declared minimal complete public response model gives

$$\begin{aligned} Q_{\mathrm{MCPR}} &= 0.6666644634090389, \\ \rho/a &= 0.7071044442750720, \\ \frac{\rho/a - 1/\sqrt{2}}{1/\sqrt{2}} &= -3.3049 \times 10^{-6}. \end{aligned}$$

The response dimensions, path table, amplitude, phase, and determinant exponent are target-informed inputs. The numerical response is therefore neither blind nor source-derived. The proximity of the two displayed Q values carries no significance or predictive weight.

Statement	Scope
Eq. (3)	exact algebra in the nonnegative-eigenvalue chamber
Theorem 4	exact implication conditional on the declared finite event packet
Icosahedral face fiber	exact local A_5/C_3 bundle and unordered spectrum
Tau window, Eq. (5)	exact quadratic consequence of the balance and ordering premises; registered decision rule
Physical family attachment	absent from the theorems; a premise of any physical reading
Phase and numerical ratios	balance leaves one phase controlling both scale-free ratios
MCPR numerical proximity	target-informed comparison-only diagnostic

6 Why residual symmetry does not finish the ratios

The family-shape problem can be stated in the real five-dimensional representation

$$W_5 \simeq \mathrm{Sym}^2_0(\mathbb{R}^3),$$

the traceless symmetric 3×3 matrices with action $A \mapsto gAg^T$.

Proposition 6 (Residual-stabilizer boundary)**.** *Invariance under a threefold or fivefold rotation about an axis n forces*

$$A = \alpha(nn^T - I/3),$$

which has a double eigenvalue. Invariance under a twofold rotation has a three-dimensional linear fixed locus, hence two projective parameters, and that locus admits simple spectrum.

Proof. Choose the rotation axis as the third coordinate. Decompose a traceless symmetric matrix into a planar scalar, planar spin-two anisotropy, transverse vector, and axial entry. Rotation by θ acts on the anisotropy by 2θ and on the transverse vector by θ. For $\theta = 2\pi/3$ or $2\pi/5$, neither block has a nonzero fixed vector, so only the axial combination remains. For $\theta = \pi$, the planar symmetric block survives:

$$A = \begin{pmatrix} u & v & 0 \\ v & w & 0 \\ 0 & 0 & -u-w \end{pmatrix}.$$

This space has dimension three and contains matrices with three distinct eigenvalues. □

Threefold and fivefold symmetry cannot produce three distinct masses. Twofold symmetry leaves two scale-free coordinates, exactly enough to carry the two mass ratios without selecting them. A numerical spectrum therefore requires a specific invariant potential derived from the screen dynamics. This proposition locates the missing input; it supplies no replacement fit.

7 Formal proof boundary

The algebraic core has a Lean 4 formalization. It proves the root sum, square sum, quotient formula, reciprocal-square-root identity, and the nonnegative-modulus equivalence

$$\frac{1}{3} + \frac{2}{3}r^2 = \frac{2}{3} \quad \Longleftrightarrow \quad r = \frac{1}{\sqrt{2}}.$$

The formal module deliberately does not identify signed eigenvalues with physical square roots. Positivity of all three eigenvalues, the phase selection, and the source-to-charged-family attachment are outside its statement. This division matches Theorem 1: the checked algebra is exact, and the physical reading is a separate premise.

8 Conclusion

The regular three-corner face carrier turns the Koide relation into one transparent modulus condition. In the positive chamber,

$$Q = \frac{2}{3} \quad \Longleftrightarrow \quad \rho/a = \frac{1}{\sqrt{2}}.$$

The finite tracial event packet supplies this balance conditionally through equal block weights and the canonical square-root representation. Under the balance and mass-ordering premises the measured electron and muon masses fix the tau mass inside the 72-eV window of Eq. (5), and a registered decision rule fixes which later measurements refute the balanced-circulant premise; the world-average and Belle II comparisons are retrospective. The result does not determine the phase or mass ratios and does not attach the local face fiber to physical charged leptons. Its durable content is an exact circulant identity, an exact finite conditional balance theorem, a conditional tau-mass window with a registered decision rule, and a sharp boundary between the checked algebra and the physical attachment premises.

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